

SME3006/AUS3006 Machine Learning – 2026 Fall

Introduction to ML



INHA UNIVERSITY

What is artificial intelligence?

- > An agent that can learn and make decisions through perception, reasoning and inference.

인공지능 	
인간적인 사고 ‘기계가 인간처럼 생각과 마음을 가지게 하는 것’ (하우겔란드, 1985)	합리적인 사고 ‘인지와 추론, 행동을 가능하게 하는 계산의 연구’ (원스틴, 1992)
인간적인 행동 ‘인간이 지능적으로 행동해야 하는 것을 수행할 수 있는 기계를 만드는 기술’ (벨만, 1978)	합리적인 행동 ‘인공적으로 만들어진 것의 지능적인 행동에 관련된 것’ (닐슨, 1998)
인간의 능력 기준	합리성 기준

▲ 인공지능의 여러 가지 정의

What is artificial intelligence?

- > An agent that can learn and make decisions through perception, reasoning and inference.
- > Difference between a regular air conditioner and an AI air conditioner



Acting humanly: the Turing test approach

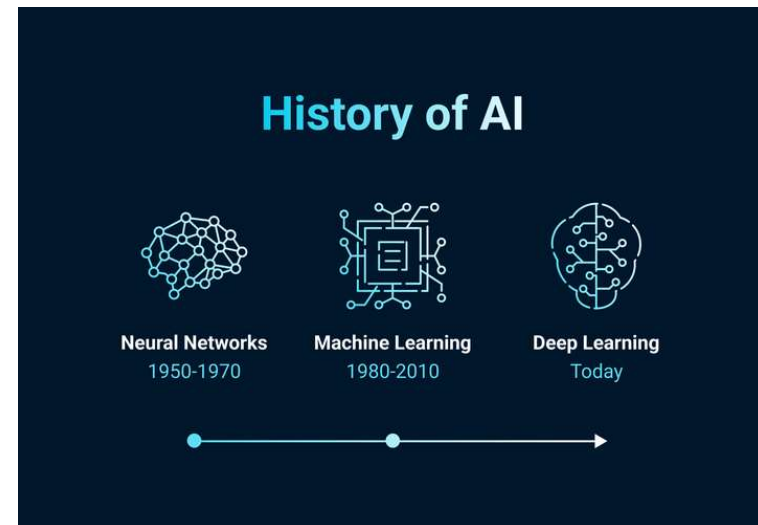
- > Can a machine think?

- > Turing Test
 - **Natural language processing** to communicate successfully in a human language
 - **Knowledge representation** to store what it knows or hears
 - **Automated reasoning** to answer questions and to draw new conclusions
 - **Machine learning** to adapt to new circumstances and to detect and extrapolate patterns

- > Total Turing Test
 - **Computer vision** and speech recognition to perceive the world
 - **Robotics** to manipulate objects and move about

History of AI

- > 1950s Perceptron
- > 1980s Back-propagation, Bayesian Network, **Decision Tree**
- > **1990s SVM, Ensemble, EM, RL**
- > **2000s Kernel Methods**
- > 2010s AlexNet, RNN, GAN
- > 2020s Transformer, LLM



What is machine Learning?

- > Extracting knowledge from data
- > Think of a spam filter
 - Make up a blacklist of words
 - Lots of 'if' and 'else' decisions to process data
- > Disadvantages
 - The logic required to make a decision is specific to a single domain and task
 - Designing rules requires a deep understanding of how a decision should be made by a human expert
- > Detecting faces in images.

What can we do with ML?

- > <https://quickdraw.withgoogle.com/>
- > First image of a black hole
- > Early detection of pancreatic cancer
- > OpenAI defeats Dota2 world champions
- > AlphaFold – prediction of a protein's 3D shape (Nobel Prize)

What can we do with ML?

Article

Highly accurate protein structure prediction with AlphaFold

<https://doi.org/10.1038/s41586-021-03819-2>

Received: 11 May 2021

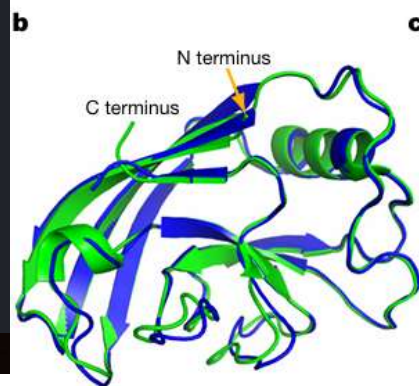
Accepted: 12 July 2021

Published online: 15 July 2021

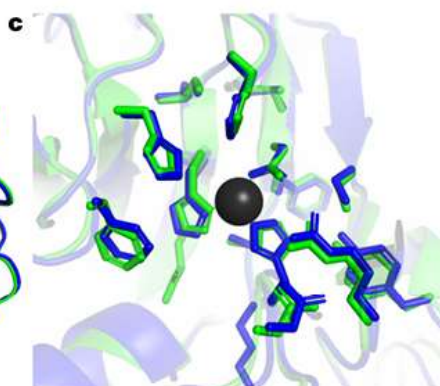
Open access

 Check for updates

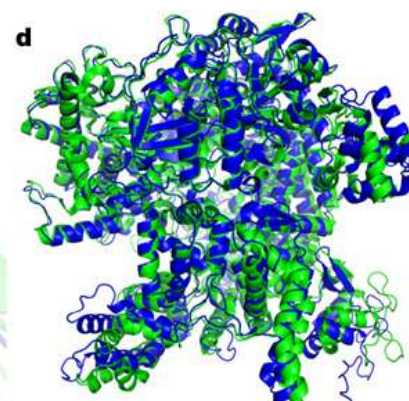
John Jumper^{1,4}, Richard Evans^{1,4}, Alexander Pritzel^{1,4}, Tim Green^{1,4}, Michael Figurnov^{1,4}, Olaf Ronneberger^{1,4}, Kathryn Tunyasuvunakool^{1,4}, Russ Bates^{1,4}, Augustin Židek^{1,4}, Anna Potapenko^{1,4}, Alex Bridgland^{1,4}, Clemens Meyer^{1,4}, Simon A. A. Kohl^{1,4}, Andrew J. Ballard^{1,4}, Andrew Cowie^{1,4}, Bernardino Romera-Paredes^{1,4}, Stanislav Nikolov^{1,4}, Rishub Jain^{1,4}, Jonas Adler¹, Trevor Back¹, Stig Petersen¹, David Reiman¹, Ellen Clancy¹, Michal Zielinski¹, Martin Steinegger^{2,3}, Michalina Pacholska¹, Tamas Berghammer¹, Sebastian Bodenstein¹, David Silver¹, Oriol Vinyals¹, Andrew W. Senior¹, Koray Kavukcuoglu¹, Pushmeet Kohli¹ & Demis Hassabis^{1,4}



AlphaFold Experiment
r.m.s.d.₉₅ = 0.8 Å; TM-score = 0.93



AlphaFold Experiment
r.m.s.d. = 0.59 Å within 8 Å of Zn



AlphaFold Experiment
r.m.s.d.₉₅ = 2.2 Å; TM-score = 0.96

0.84
0.82
0.78
0.86
0.82
0.74
0.66
0.72
0.68
0.68
0.55
0.55
0.225 0.9746 1 0.55
0.788 0.6157 0 0.55
0.883 0.4695 0 0.55
0.996 0.08716 1 0.3

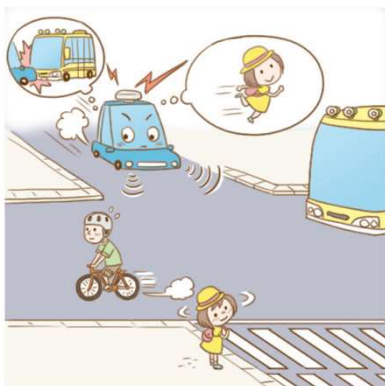
What can we do with ML?

Recommendation system



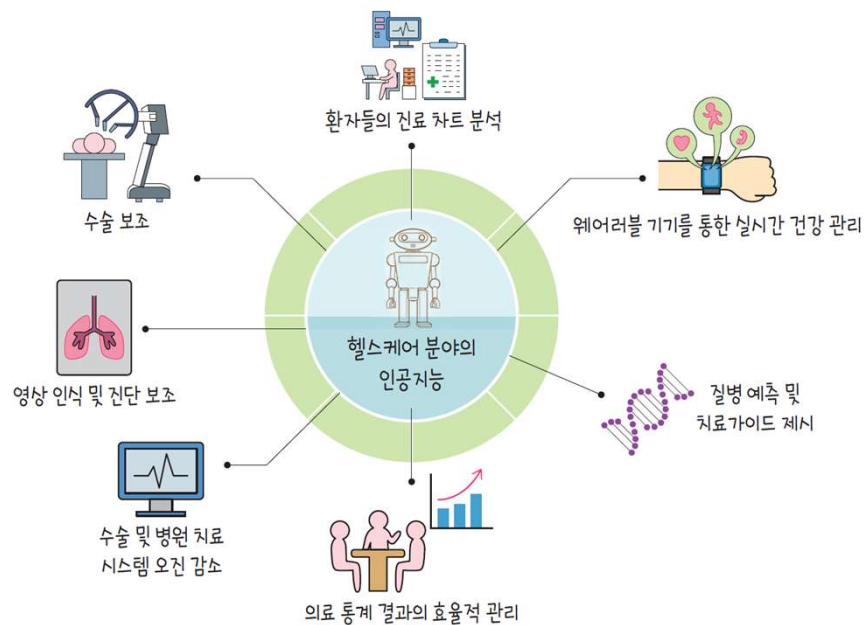
▲ 교보문고 추천 시스템

Self-driving cars



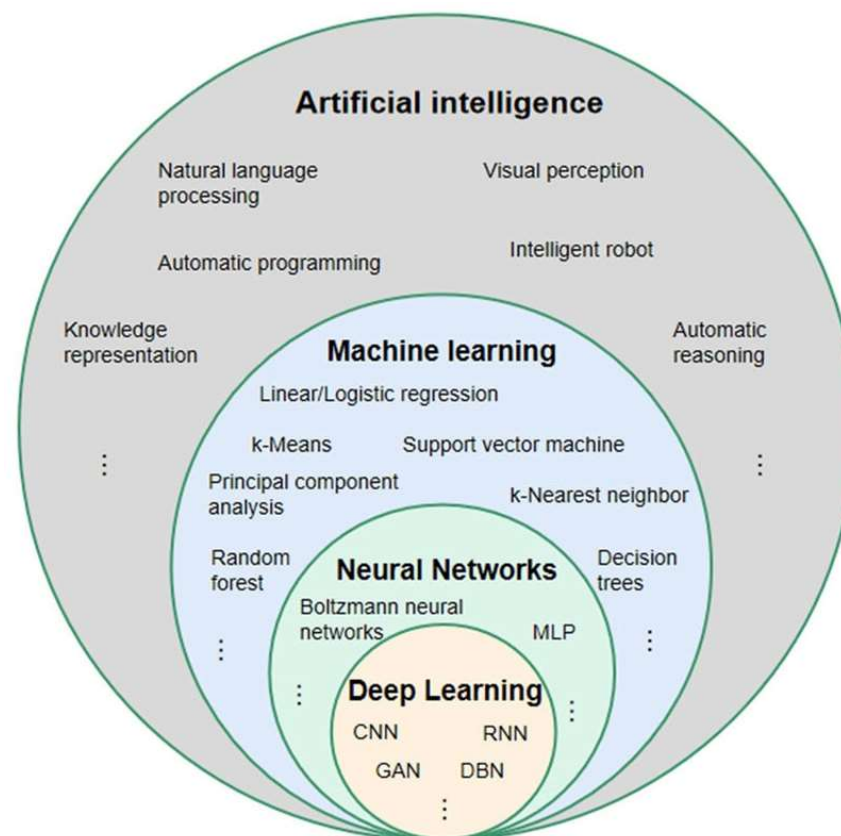
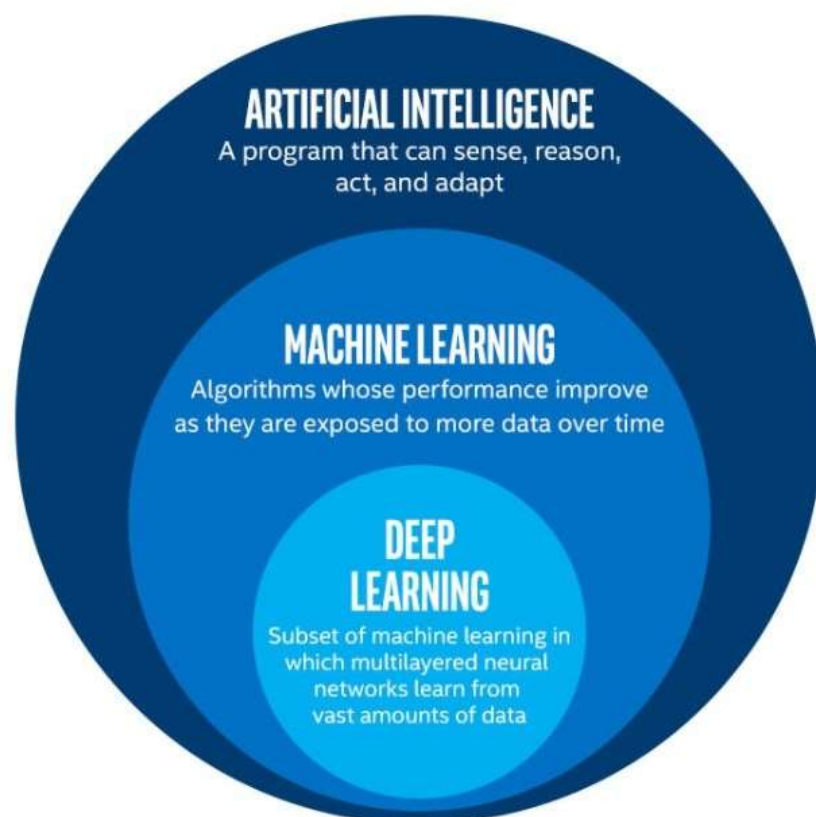
▲ 자율 주행 자동차

Healthcare and medical treatment



▲ 의료와 헬스 케어 분야의 인공지능 기술

Level of AI

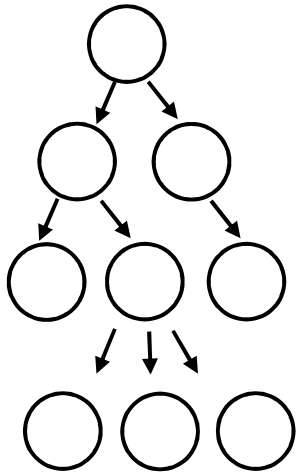


Level of AI

Artificial Intelligence

Symbolic AI

If then ..

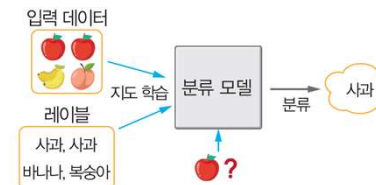


PDDL
Prolog
Expert Systems

Machine Learning

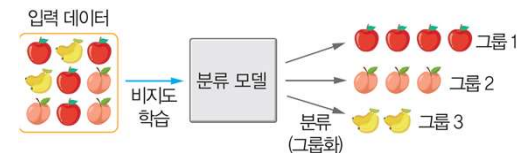
Supervised Learning

We have labels



Unsupervised Learning

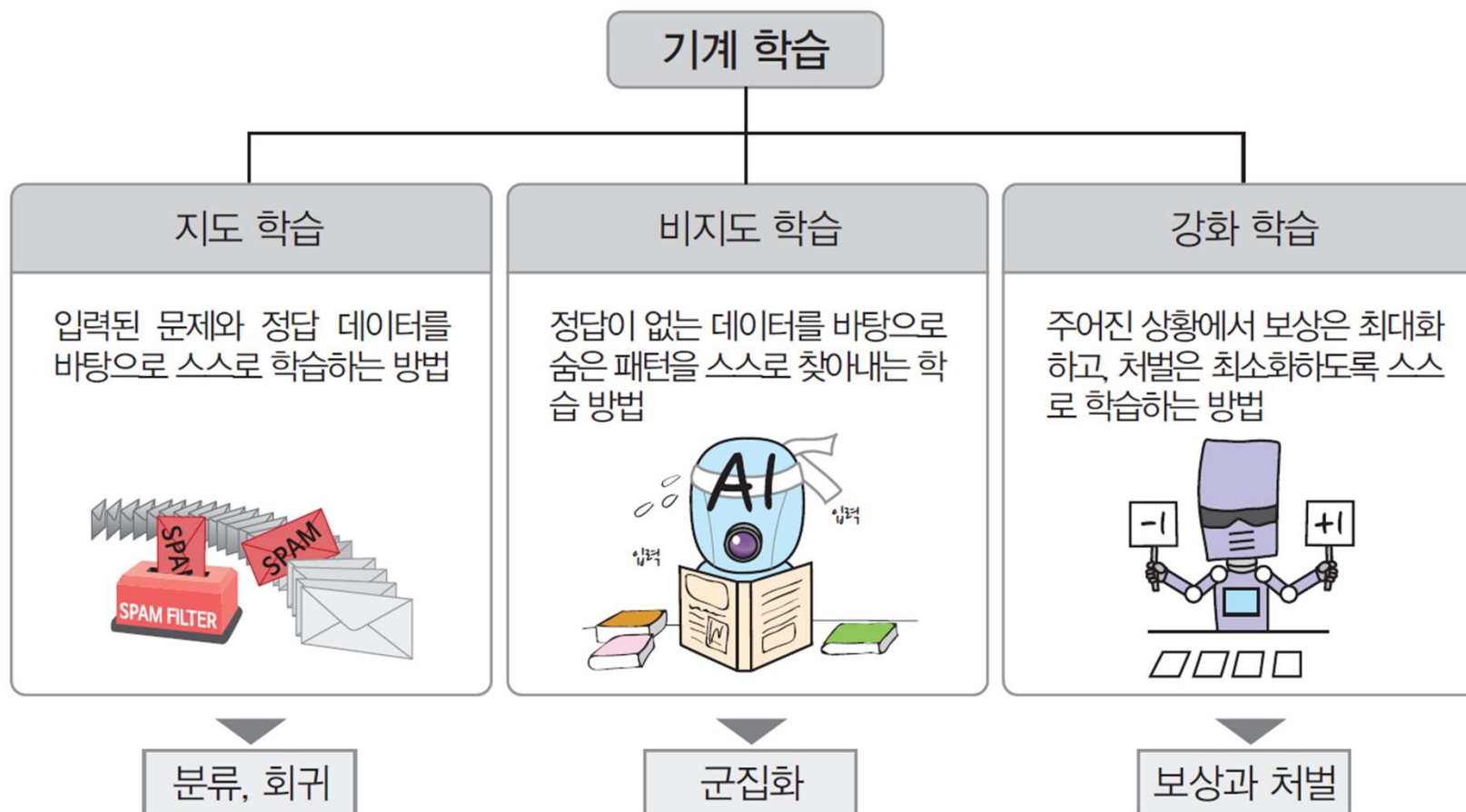
No labels



Reinforcement Learning

Learn from experience

Types of ML



Examples of learning problems

- > Recognizing digits in images
- > Identifying tumor in an x-ray
- > Classifying email as spam or not
- > Diagnosing disease from symptoms
- > Predicting the price of a stock 6 months from now
- > Predicting selling price for individual houses
- > Netflix problems – predict rating of a movie by a customer
- > Determining credit-worthiness for mortgages or credit cards
- > Predicting 3D forces on a molecule from molecular formula

Types of learning problems

- > Classification problems – output is a label from a finite set
- > Regression problems – output is real-valued (continuous)
- > Ranking problems – output only ranks examples relative to one another
- > Unsupervised problems – such as ChatGPT, often used for generative AI

Syllabus

Week 1	Introduction to ML	Week 9	Clustering (online)
Week 2	Data preprocessing / visualization	Week 10	Project milestone presentation
Week 3	Linear and logistic regression	Week 11	Decision trees and ensemble methods
Week 4	Optimization and Gradient descent	Week 12	Uncertainty quantification and Gaussian Process
Week 5	Project proposal presentation	Week 13	Expectation Maximization and GMM
Week 6	Dimension reduction and PCA	Week 14	Gradient-free optimization
Week 7	SVM	Week 15	Final project presentation
Week 8	Mid-term		

Lectures will be..

- > Lecture recordings are provided on YouTube.
- > Language: Python
- > Env. recommendation: Google Colab or Anaconda

Prerequisite

- > Probability theory
- > Linear algebra
- > Python

Lecture plan

> 3+ Assignments

- TA: 이상현 (sh000819@inha.edu)

> Grade

- Attendance – 5% (No grade if absences reach one-quarter of total class days.)
- Assignment 15%
- Mid-term 25%
- Final-project 55%

Recommended reading

> 수학과 함께하는 고교 AI 입문

- <https://www.ebssw.kr/info/intrcn/infoTchmtrHeaderView.do?bookCode=5&tabType=AI>

Reference

> Book

- [1] 알고리즘 중심의 머신러닝 가이드, Stephen Marsland
- [2] 머신러닝 교과서 파이토치편, Sebastian Raschka
- [3] AI Modern Approach, Stuart Russel

> Lecture Notes

- [4] Stanford CS229, Andrew Ng,
- [5] UC Berkeley, Concise Machine Learning, J.R. Schewchuk
- [6] NYU Machine Learning: a lecture note, Kyunghyun Cho